

Ratchet: Intelligent Local Privacy Shield for AI Workflows

Track: Developer Tools / AI Safety & Privacy

Short Description: Ratchet is an intelligent local privacy shield that automatically detects, redacts, and restores sensitive data in real time whenever users interact with AI models or LLM applications.

Problem Statement

As AI tools and large language models become central to daily workflows, users frequently paste sensitive information—including Personally Identifiable Information (PII), credentials, internal project details, and financial figures—into external AI prompts. Once transmitted, this raw sensitive data risks being logged, reviewed by third parties, or integrated into model training datasets, causing severe compliance and data leak concerns.

Target Users

- **Developers & Engineers:** Sharing code snippets containing API keys, private tokens, or internal infrastructure details.
- **Enterprise & Knowledge Workers:** Processing customer records, financial reports, or legal contracts using generative AI.
- **Privacy-Conscious Individuals:** Seeking to leverage AI tools without surrendering personal data control.

What Ratchet Does

- **Detect & Swap:** Before your prompt leaves your computer, Ratchet scans the text to find sensitive information. It instantly swaps out private details with simple, neutral placeholders like "Name 1" or "Email 1".
- **Safe AI Processing:** The external AI model receives only the sanitized version with placeholders. It processes your request and provides a response without ever seeing your actual private information.

- **Automatic Restoration:** As soon as the AI sends the answer back to your device, Ratchet replaces the temporary placeholders with your original details so you see the complete, correct answer seamlessly.

Technical Approach & Architecture

How Reversible Redaction Works

Ratchet performs all detection, redaction, and restoration entirely on your local machine. No original text, secret mapping keys, or sensitive values are ever sent to external cloud servers or stored online. Your raw private data never leaves your device.

1. Layered Detection Pipeline

To ensure no private information slips through, Ratchet uses a multi-layered approach to detect sensitive details:

- **Pattern Matching (Regex):** Instantly catches structured data with clear formats, like credit card numbers, email addresses, IP addresses, and phone numbers.
- **Local Machine Learning (NER):** Uses lightweight, local AI models (Named Entity Recognition) to identify context-based data like human names, organization titles, and physical addresses.
- **Custom User Rules:** Allows you to specify project-specific terms, internal codenames, or custom secret keywords that should always be protected.

2. Utility Preservation

Traditional redaction tools often block out entire sentences or use black bars, making text unreadable and useless for AI models. Ratchet uses consistent placeholder labels (like "Person 1" and "Person 2") that preserve the grammar, logical structure, and relationships in your message. This ensures the AI model can still give accurate, high-quality answers.

Evaluation & The Kaizen Method

Ratchet follows the Kaizen philosophy of continuous, incremental improvement. Whenever new types of sensitive patterns, edge cases, or novel AI formats appear, the detection mechanisms are updated locally to continuously refine accuracy and lower false positives without compromising user privacy.

Real-World Impact

- **Zero-Trust AI Adoption:** Enables organizations to adopt cloud AI tools with confidence, ensuring zero sensitive data leaks.
- **Compliance Alignment:** Helps meet strict regulatory requirements (GDPR, HIPAA, SOC 2) without restricting AI productivity.
- **Frictionless Experience:** Delivers automated, transparent protection directly within standard browser and desktop workflows.

Tech Stack

- **Core Engine:** Python / Rust (High-performance local text processing)
- **NLP & NER:** spaCy / ONNX Runtime (Lightweight local machine learning)
- **Interface & Extensions:** TypeScript, React, WebExtensions API

Submission Links

- **GitHub Repository:** <https://github.com/example/ratchet>
- **Demo Video:** <https://youtube.com/watch?v=example>
- **Live Project / Extension:** <https://ratchet.example.com>

Why Ratchet is Different

- **Reversible, Not Destructive:** Unlike standard tools that delete private data permanently, Ratchet restores your data back into the AI's answer.
- **Zero Cloud Reliance:** Security rules and redactions run locally without requiring remote security proxies or third-party cloud servers.
- **Seamless Workflow:** Operates automatically in the background without breaking your chat experience or requiring manual copy-pasting of placeholders.

Handling Weaknesses & Edge Cases

No detection system is 100% perfect on its own. When ambiguous or complex text makes detection difficult, Ratchet handles weaknesses through smart fallback strategies:

- **Contextual Fallbacks:** If an entity is uncertain, Ratchet errs on the side of caution and redacts it to prevent accidental leaks.

Interactive Review: Provides visual indicators allowing you to manually highlight or un-redact specific words before sending your request.